

Medical Imaging Dataset Exploration

Jovita Bhasin

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1 Introduction

Medical imaging datasets are widely used for developing machine learning and deep learning models for disease diagnosis and medical decision support. In this report, two popular medical imaging datasets are explored: the NIH ChestX-ray14 dataset and the HAM10000 Skin Lesion dataset. For each dataset, information regarding the imaging modality, dataset size, labels, imbalance, and challenges is discussed.

2 Dataset 1: NIH ChestX-ray14

2.1 Type of Imaging Data

The NIH ChestX-ray14 dataset consists of chest X-ray (radiography) images collected from patients for the detection of thoracic diseases.

2.2 Number of Images

- 112,120 chest X-ray images
- 30,805 unique patients

2.3 Available Classes/Labels

The dataset contains 14 disease labels:

- Atelectasis
- Cardiomegaly
- Effusion
- Infiltration
- Mass
- Nodule
- Pneumonia
- Pneumothorax

- Consolidation
- Edema
- Emphysema
- Fibrosis
- Pleural Thickening
- Hernia

Additionally, images may be labeled as *No Finding*, indicating the absence of detectable disease.

2.4 Dataset Imbalance

The dataset is significantly imbalanced. Common findings such as Infiltration and Effusion contain many more samples than rare conditions such as Hernia. The *No Finding* category also contains a large number of images.

2.5 Challenges

- Class imbalance among disease categories.
- Multi-label classification since one image may contain multiple diseases.
- Labels were automatically extracted from radiology reports and may contain errors.
- Variations in image quality and patient positioning.

2.6 Brief Summary

The NIH ChestX-ray14 dataset is one of the largest publicly available chest radiography datasets. Its large size makes it useful for training deep learning models for automated disease detection. However, label noise, class imbalance, and multi-label annotations make the dataset challenging for machine learning applications.

3 Dataset 2: HAM10000 Skin Lesion Dataset

3.1 Type of Imaging Data

The HAM10000 dataset consists of dermatoscopic images of pigmented skin lesions used in dermatological diagnosis.

3.2 Number of Images

- 10,015 dermatoscopic skin lesion images

3.3 Available Classes/Labels

The dataset contains seven diagnostic categories:

- AKIEC (Actinic Keratoses and Intraepithelial Carcinoma)
- BCC (Basal Cell Carcinoma)
- BKL (Benign Keratosis-like Lesions)
- DF (Dermatofibroma)
- MEL (Melanoma)
- NV (Melanocytic Nevi)
- VASC (Vascular Lesions)

3.4 Dataset Imbalance

The dataset is highly imbalanced. The NV (Melanocytic Nevi) class contains the majority of samples, while classes such as DF and VASC contain relatively few images.

3.5 Challenges

- Severe class imbalance.
- Similar visual appearance among different lesion categories.
- Variations in lighting, skin tone, and image acquisition devices.
- Difficulty in accurately identifying lesion boundaries.

3.6 Brief Summary

HAM10000 is a benchmark dataset for skin lesion classification and melanoma detection. It is widely used for developing computer-aided diagnostic systems. Despite its usefulness, the dataset presents challenges due to severe class imbalance and visual similarity among lesion types.